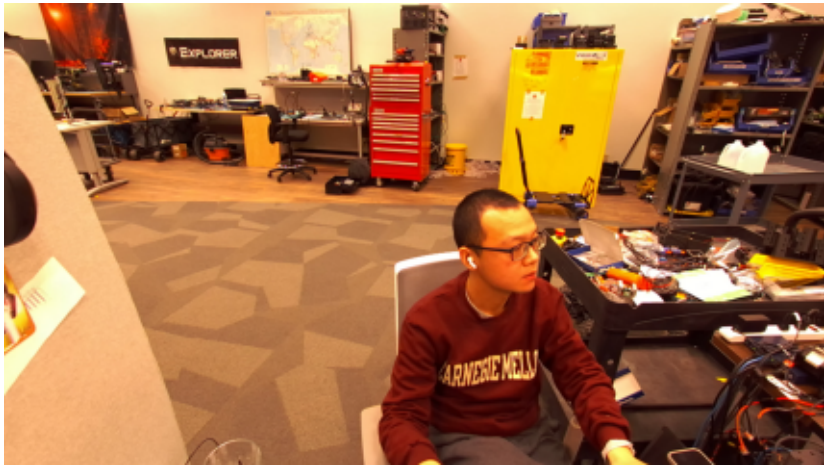
ZED depth: ONNX vs PyTorch

FAIL: outputs differ beyond tolerance

Model: neural_depth_5.4.onnx | atol=0.003, rtol=0.003

Left RGB input: im0.png



Right RGB input: im1.png



Output	Mean abs.	Max abs.	RMSE	P99 abs.	Outside tol.	Result
disparity	0.00885754	0.73699	0.0307165	0.143136	5.624%	FAIL
disparity_right	0	0	0	0	0%	PASS
confidence	0.00178466	0.149481	0.00515589	0.0230319	12.8%	FAIL

Input HxW: [621, 1104] | network HxW: [256, 512] | bottom/right padding: [0, 0]

Output shape: [1, 1, 128, 256] | comparison includes padding; disparity is not metric depth.

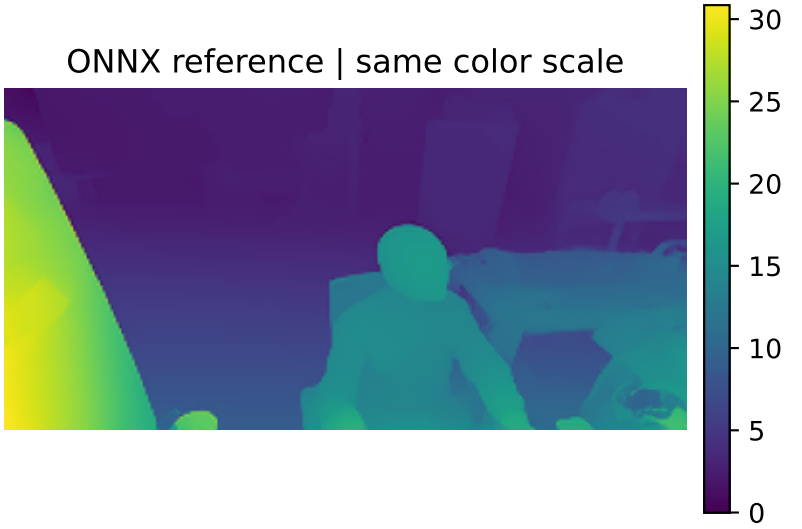
Reference: ONNX Runtime standard layers + independent NumPy custom operators. No PyTorch tensors are injected.

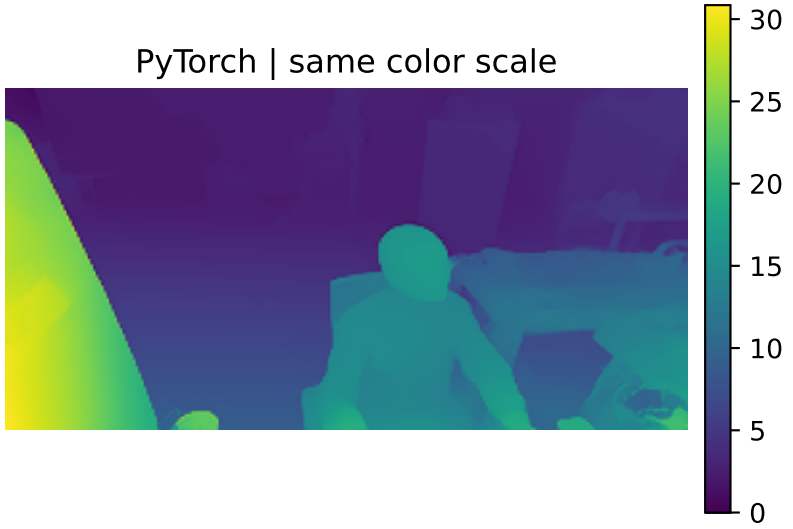
disparity_right is a zero-filled placeholder; matching zeros does not validate right-view depth.

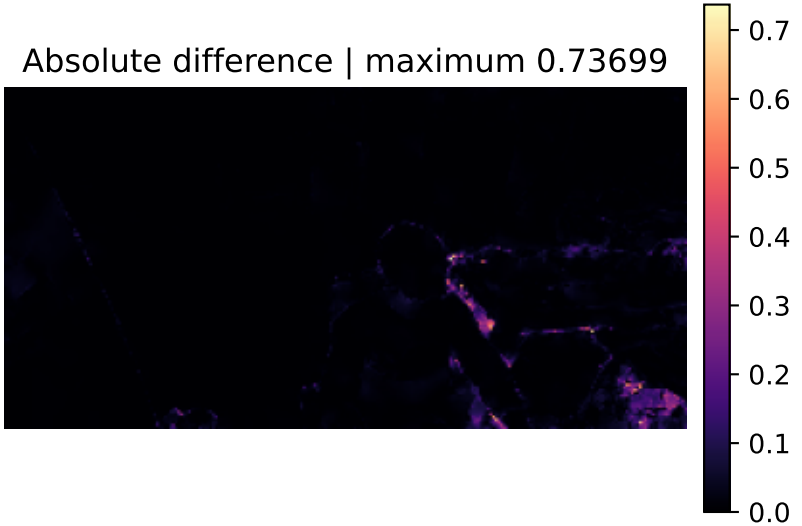
This measures implementation agreement on one stereo pair. No ground truth or native ZED CUDA parity was tested.

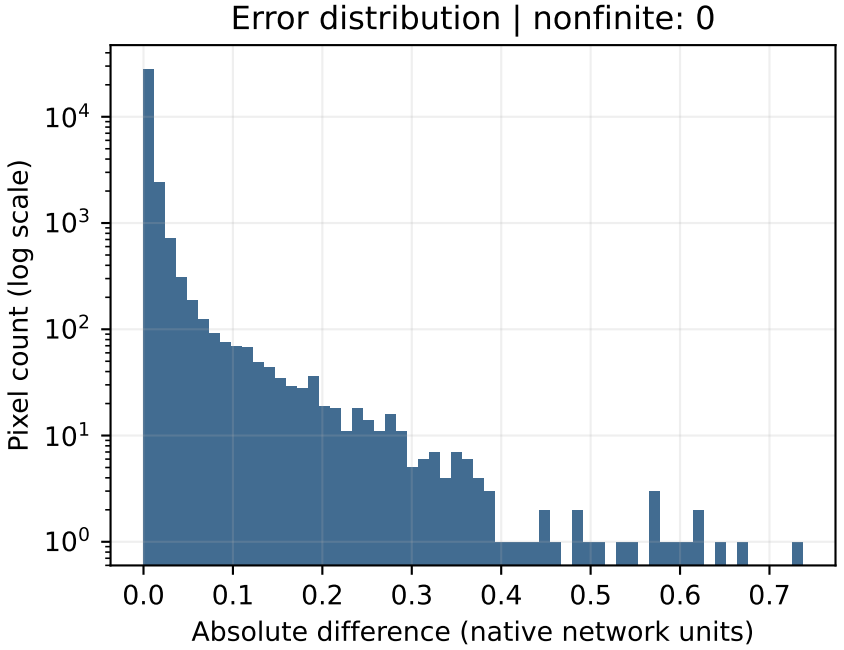
Timing: PyTorch forward 0.142s; reference setup + run 0.322s.

Disparity output analysis



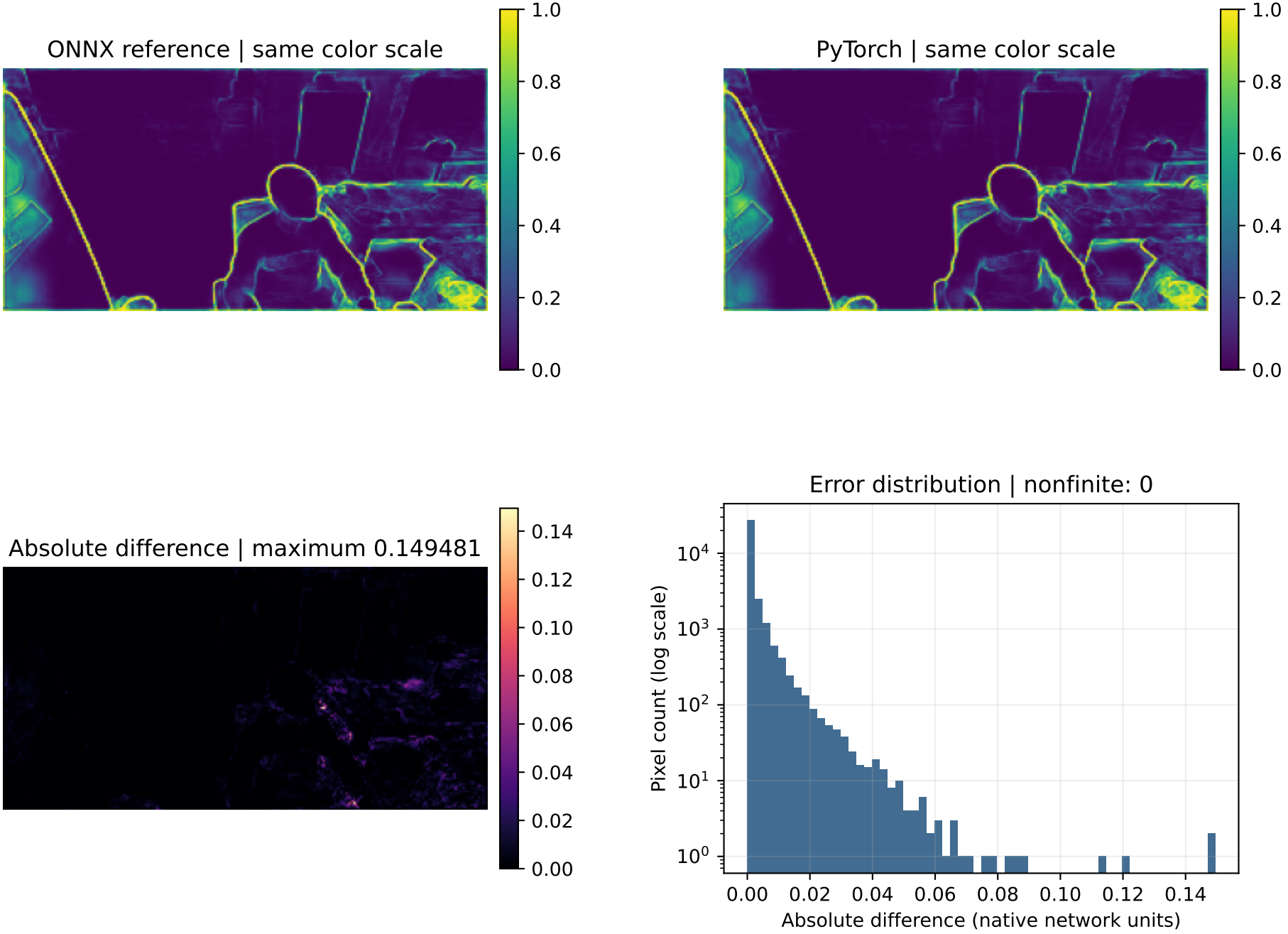






Out-of-tolerance: 5.62439% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.

Confidence output analysis



Out-of-tolerance: 12.8021% | Difference colors are magnified to the maximum error; inspect numeric values, not color alone.